

CICESE
Foundations of Deep Learning
Assignment 2
Due date: March 4, 2026

Instructor: Jesús Guillermo Falcón-Cardona

General instructions: Please read the specific instructions in each section carefully (and more than once, if necessary). Eligibility for evaluation strictly requires the submission of: (i) the implementation, (ii) a clear description of the algorithmic components, and (iii) a discussion of the results. Omission of any of these elements will result in a zero grade.

The description of the algorithmic components and the discussion of the results must be written in a scientific and academic style, supported by clear, quantitative, and rigorous arguments. Superficial or vague explanations will also result in a zero grade. When presenting the results, do not merely restate what is evident from the tables or figures. Instead, provide an in-depth analysis that goes beyond the obvious, and explain why the algorithms behave as they do. This is more valuable for the instructor and your grading.

Reports may be written in Spanish; however, submissions in English are preferred. Regardless of the language, the report must be grammatically correct and well-written, as this is expected at the graduate level.

If you have doubts about the Assignment, do not hesitate to contact the Instructor as soon as possible. Good luck.

1 Convolutional Neural Networks and PyTorch Utilization

Main goal: The goal of this assignment is to evaluate your understanding of:

- The mathematical foundations of CNNs.
- Regularization techniques (weight decay, dropout, early stopping).
- Normalization methods (BatchNorm and LayerNorm).
- Architectural design principles (e.g., ideas behind AlexNet, VGG, ResNet, GoogLeNet).

- Implementation and experimentation in PyTorch.
- This is not a “train and report accuracy” task. You are expected to design, justify, experiment, and critically analyze.

1.1 Part 1: Architectural Design and Theoretical Justification

Based on your current knowledge and understanding of CNNs, design your own CNN to be trained on the CIFAR-10 dataset. Your design choices must be reported based on the following questions.

1. Kernel sizes: What sizes do you use and why?
2. Network depth: Why is your model that deep?
3. Will you include residual connections inspired by ResNet? Justify.
4. Where do you place normalization layers (before or after activation)? Why?
5. What activation function do you use and why?
6. Estimate the total number of parameters and discuss model capacity.
7. What type of inductive bias does your architecture impose?
8. Provide a diagram of your architecture.

In case you have a GPU, train your model on CIFAR-10. Otherwise, use MNIST while only using CPU processing.

1.2 Part 2: Experiments Using Your CNN

Train your model on the selected dataset under the following conditions, using mini-batch SGD:

1. No regularization.
2. Weight decay regularization (i.e., L^2 regularization).
3. With dropout.
4. With weight decay + dropout + early stopping.

For each condition, report the following:

- Training and validation loss curves.
- Training and validation accuracy curves.
- Final test accuracy and confusion matrix.

- Generalization gap.
- Convergence behavior.

Additionally:

- Compare Batch Normalization vs Layer Normalization within the same architecture.
- Keep all other hyperparameters fixed when comparing.

You must explain the observed differences, not just report numbers.

1.3 Part 3: Analytical Discussion

Answer the following questions:

1. Is your model underfitting or overfitting? How do you know?
2. How does weight decay affect the norm of the weights?
3. How does dropout affect gradient flow?
4. Does BatchNorm improve optimization stability in your experiments? Why might that be?
5. Philosophically, is your architecture closer in spirit to AlexNet, VGG, or ResNet, or others? Explain.

This section evaluates your depth of understanding.

1.4 Part 4: Engineering Constraint

Design a model that fulfills one of the following constraints:

1. Maximum 500k parameters.
2. Maximum 5M FLOPs.
3. Training under 5 minutes.

You must justify how you met the constraint. Then compare the performance of this model against the best one from the previous experiments. Provide numerical data that supports the comparison.

2 Jupyter Notebook Structure

Submit your homework as a Jupyter Notebook, ensuring that the results of all experiments are saved. For each experiment, describe the algorithmic components and provide a discussion of the results next to the corresponding elements.

Moreover, submit all additional files required to execute your work if the Instructor requires re-executing your script. Be sure that all your programs are bug-free and can be executed without any problem.